DETAILED DESCRIPTION OF THE INVENTION — DEVICE GEOMETRY VARIATIONS AND ANATOMICAL FIT ARCHITECTURES

The intraluminal physiologic monitoring device disclosed herein may be implemented using a wide variety of geometric configurations intended to optimize comfort, positional stability, sensing accuracy, and manufacturability across different anatomical environments.

In various embodiments, the insertable body may include cylindrical geometries, tapered probes, bullet-shaped bodies, capsule configurations, conical transitions, curved shafts, segmented structures, or asymmetric forms designed to accommodate anatomical variation. Cross-sectional shapes may include circular, oval, elliptical, polygonal, flattened, or adaptive geometries capable of deforming under physiologic pressure.

Certain embodiments may employ variable-diameter regions along the device length to support gradual insertion, enhanced retention, or localized sensing concentration. Transitional contours may reduce insertion resistance while improving tissue contact consistency during measurement.

Expandable embodiments may include inflatable sections, elastically deformable structures, shape-memory materials, or mechanically actuated expansion mechanisms configured to conform to surrounding tissue geometry. Expansion may be uniform, localized, segmented, or dynamically controlled.

Flexible or articulated embodiments may incorporate bendable cores, segmented joints, compliant hinges, or soft robotic structures allowing the device to conform to curved anatomical pathways while maintaining sensing functionality.

Retention geometries may include flanges, wings, compliant bulbs, textured surfaces, ridges, or anatomically adaptive profiles configured to stabilize device orientation. Retention structures may be symmetrical or asymmetrical depending on intended positioning behavior.

In certain implementations, the device may include interchangeable outer shells, size variants, modular extensions, or customizable geometries allowing adaptation to individual users or clinical requirements. Geometry selection may be determined through calibration routines, user selection, or automated fitting algorithms.

Surface topology may also vary and may include smooth finishes, micro-textured surfaces, lubricious coatings, compliant overmolds, or patterned structures designed to improve comfort, reduce friction, or enhance sensing reliability.

The geometric variations described herein are illustrative rather than exhaustive. Any device geometry capable of supporting intraluminal placement and physiologic sensing is considered within the scope of the invention, including future geometries enabled by advances in materials science, manufacturing methods, or adaptive structural technologies.